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Obstacle Avoidance of Robotic Arm in Path Planning Based on RRT Algorithm

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***Abstract*—This paper delves into the obstacle avoidance issue of robotic arms in dynamic environments. In modern industry, robotic arms are applied widely and used to improve efficiency. However, during the application, there are numerous issues that robotic arms expose. Aiming at addressing several challenges of robotic arm in the industrial scenario, it is urgent to explore an effective algorithm for obstacle avoidance. The paper introduces and discusses the RRT algorithm and the optimized RRT smooth algorithm. Through the simulation, several obstacles such as cube, sphere and the combination of both of them are introduced in the simulation space to verify the effectiveness and efficiency of the RRT smooth algorithm. The graphs from the simulation of MATLAB demonstrates that the performance of RRT smooth is better than RRT algorithm to some extent and there are some visible results to be compared such as the smoothness and the efficiency. Specifically, the RRT smooth algorithm not only decrease the average path length and processing time, but also decrease the maximum velocity and acceleration as well as the average acceleration, while ensuring the high average velocity. The results demonstrate that the RRT smooth algorithm effectively optimize the motion performance of the robotic arm and is helpful for extending the lifespan of the robotic arm by reducing the vibration during the movement. Although RRT smooth is better, there still exists some defects in the algorithm such as blindness and randomness and the generated path is not the best.**

Keywords-path planning; RRT algorithm; RRT smooth algorithm; obstacle avoidance; robotic arm

I. Introduction

In recent years, robotic arms are widely applied in various production scenarios as they can carry out repeated and precise tasks with the same links as human arms with a wrist, elbow and a shoulder[1], which entirely transforms the production models of manufacturing, logistics and healthcare. However, there are numerous challenges on the actual motion of robotic arms and the most noticeable issue is the obstacle avoidance that the robotic arm operating in the dynamic environment may face. Previous studies have extensively explored robot path planning. In static environments, algorithms such as A*[2] and the artificial potential field method[3] have proven highly effective in finding optimal paths. These algorithms employ graph search techniques, which represent positions in the workspace as nodes and possible movements between nodes as edges through calculating the traversal cost of each edge and using heuristic functions to obtain the shortest path. For example, in material handling operations for static factory layouts, the A* algorithm can efficiently guide the robotic arm to the target object position. However, in dynamic environments, these algorithms exhibit significant limitations. The static nature of their search space and the assumption that obstacles remain unchanged make them difficult to adapt to scenarios with moving obstacles, which may result in collisions or require frequent recalculation that are both time-consuming and resource-intensive.

The RRT algorithm was proposed by Steven M. LaValle in 1998[4] and is considered revolutionary in path planning in high-dimensional spaces and complex obstacle environments. This algorithm solves the limitations of traditional algorithms in dynamic environments by randomly sampling points in the search space and gradually building a tree-like structure. This approach enables it to quickly explore large areas and find feasible paths even in the presence of complex geometries and moving obstacles. For instance, in the scenario of unmanned aerial vehicle (UAV) path planning, when a UAV needs to avoid other aircraft or birds and other moving obstacles, RRT can rapidly generate paths to bypass these dangers. However, the paths generated by the RRT algorithm often have the problem of non-optimality. Due to the influence of the random sampling process, these paths are usually longer and contain unnecessary twists and turns. In that case, numerous variants of RRT have been proposed. Among them, there is an improved RRT algorithm for dual-arm robots, which utilized a goal probability bias and cost function in the RRT algorithm[5].

This study emphasizes on developing and improving efficient obstacle avoidance algorithms for robotic arms in dynamic environments. The research investigates the RRT algorithm and its optimized variant, aiming to improve path planning performance through simulations and experiments with the robotic arm. The designed obstacles are a cube and a sphere. The length, width and height of the cube are respectively 200, 30 and 100 while the radius of the sphere is 40. The robotic arm seeks the path from the given start point and the target point through RRT algorithm and RRT smooth algorithm. The major indexes of estimation including path length and processing time show the efficiency of the two algorithms.

The paper is structured as follows: Section 2 outlines the methodology, including the process of the experiment and RRT algorithm. Section 3 presents the path optimization through utilizing RRT smooth and Section 4 discusses the visible results of the simulation. Ultimately, Section 5 is the conclusion of the whole paper.

II. Methodology

A. The process of the experiment

The robotic arm was employed in the MATLAB as the simulation platform, which was equipped with sensors to

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monitor real-time position and attitude data. The task of the robotic arm is to reach the target point from the start point while avoiding the static obstacle that was introduced to simulate a realistic working environment. When setting up obstacles, a cube and a sphere were chosen to be the obstacle. The cube has dimensions of 200 in length, 30 in width and 100 in height, while the sphere has a radius of 40. These sizes and shapes of obstacles simulate various types of barriers that the robotic arm may face in real industrial scenarios, which provides a diverse environment for testing obstacle avoidance performance of the algorithm. In actual industrial production, robotic arms might encounter equipment and material piles with different shapes and sizes. Choosing cubes and spheres can more comprehensively simulate these scenarios. As shown in Fig. 1, the process of the experiment illustrates the specific steps of the experiment. First, select the PUMA560 robotic arm in MATLAB as the research object and build the robot arm model. Then, set the positions of obstacles legitimately within the workspace. Next, choose RRT the algorithm and RRT smooth algorithm as path planning algorithms. Subsequently, control the robotic arm to move from the starting point to the target point while sensors collecting real-time position and posture data of the robotic arm during movement. Finally, display the motion trajectory of the robotic arm and record relevant data for subsequent evaluation of algorithm performance. Throughout the experiment, multiple repeated experiments are conducted to ensure the reliability and stability of the experimental results. After each experiment, the recorded data is analyzed to observe the performance of different algorithms under various experimental conditions, which provides a foundation for optimizing and improving the algorithms. Table I demonstrates the standard D-H parameters sheet of PUMA560 robotic arm, which is used for simulation. These parameters precisely define the geometric and motion characteristics of each joint in the robotic arm, which is the pivotal factor in ensuring the accuracy of the simulation. For example, α_1 = - pi/2, $a_1 = 0$, $d_1 = 0$ and $\theta_1 = 0$. These parameters determine the initial position and range of motion of joint L1 in the simulation space.

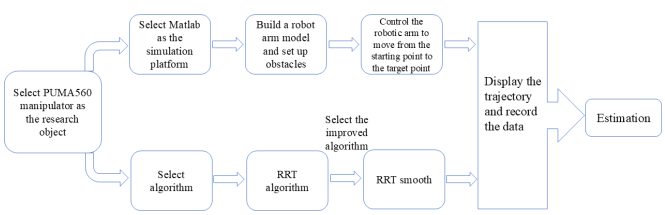

Figure 1. The process of the experiment

TABLE I THE STANDARD D-H PARAMETER SHEET

Joint	Parameters			
	α_i	a_i	d_i	θ_i
L1	-pi/2	0	0	0
L2	0	0.432	0	0
L3	-pi/2	0.02	0.149	0
L4	pi/2	0	0.433	0
L5	-pi/2	0	0	0
L6	0	0	0	0

B. RRT algorithm

The Rapidly-exploring Random Tree (RRT) is a sampling-based algorithm for path planning, which is suitable for high-dimensional spaces. This algorithm is primarily used to solve problems in areas such as robot path planning and unmanned aerial vehicle path planning, especially in the complex environments. The RRT algorithm explores the search space by randomly sampling points and gradually constructing a random tree. RRT algorithm steps:

1) Define the starting point as the root node of the random tree.

2) Randomly select a point q_{rand} in the search space.

3) Find the node q_{near} in the existing random tree that is closest to q_{rand}.

4) Extend a fixed step from q_{near} towards q_{rand} to obtain a new node q_{new}.

5) Check whether the path from q_{near} to q_{new} collides with any obstacles. If there is no collision, add q_{new} to the random tree.

6) Repeat steps 2 - 5 until a node in the random tree is close to the target point or the maximum number of iterations is reached.

RRT can quickly seek a feasible path in complex environments but has strong blindness and randomness. The paths generated are usually not optimal, with many twists and unnecessary turns, resulting in relatively long path lengths.

C. Path optimization

Aiming to generate smoother path, RRT smooth algorithm is proposed to be the optimized method.

The RRT smooth algorithm checks the feasibility of connections between adjacent points on the path and removes intermediate points that cause sharp turns. Specifically, for any two points q_i and q_j on the path, it checks whether the straight - line connection from q_i to q_j collides with obstacles. If there is no collision, these two points can be directly connected, thus removing intermediate nodes and making the path more concise and smoother.

Additionally, in the path planning, the change in joint angles is also crucial besides the geometric shapes of the path. The RRT smooth algorithm employs polynomial trajectory planning (such as cubic polynomials) to generate intermediate transition values for joint angles, which makes the change in joint angles between adjacent path points smoother and more continuous that avoid sudden changes in joint motion, reduce the impact and vibration on robot movement and improve the smoothness and efficiency when seeking the path. This polynomial-based approach has been shown to effectively reduce the impact and vibration during movement, as demonstrated in relevant research [6].

The RRT smooth algorithm generally shortens the total path length through removing unnecessary nodes that could reduce the processing time and energy consumption in practical applications, which improves the efficiency of robot work. It is similar to the approach described in the work of [7].

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III. Results and discussion

The result presented in this section is based on 50 carried out experiments, from which the suitable calculated step size and threshold are 9 and 1 for both RRT algorithm and RRT smooth algorithm.

Under these conditions, the mean time, average path length and whether reach the target point are calculated from simulation in MATLAB as shown in Table II. On one hand, the average path length generated by RRT smooth is 254.45, which is shorter than the path length of 292.33 generated by RRT. On the other hand, RRT smooth takes less processing time to reach the target point (only 0.98 seconds), while the RRT algorithm requires 1.65 seconds. The same result is calculated by Zhu Ying et al.[8] and Ding zhijie et al.[9]. The mean time and average path length of their optimized algorithm are both less than the original RRT algorithm. From Table II, it is obvious that the mean time and average path length of RRT smooth is less than RRT, which means RRT smooth is a feasible algorithm to improve the performance of the robotic arm to some extent based on the same condition and the successful arrival. These results are significant for applications of the robotic arm in industrial production. A shorter path length is related to less energy consumption and the robotic arm generally is required to operate the repeated tasks. In that case, the robotic arm may operate in a short distance, which could decrease the cost of energy consumption in the long term, especially in a large-scale manufacturing plants where require multiple robotic arms to operate simultaneously. Therefore, the cumulative energy saving could be prominent, contributing to more sustainable and cost-effective production processes.

TABLE II Comparison of the simulation result between algorithm

Algorithm	Performance		
	Mean Time(s)	***Average path length***	***Reach the target point or not***
RRT	1.65	292.33	Yes
RRT smooth	0.98	254.45	Yes

Table III shows the maximum velocity and acceleration of six joints during the motion by RRT and RRT smooth of the cube obstacle. As shown in the Table III, the maximum velocity and acceleration data generated by RRT smooth are generally smaller than RRT, which demonstrates that the RRT smooth algorithm could effectively decrease the maximum velocity and acceleration, making the motion more stable and eliminating the vibration of the robotic arm to some extent. For example, the maximum velocity and acceleration of joint L1 are 0.129 and 0.048 with RRT algorithm, but 0.11 and 0.024 with RRT smooth algorithm, which is similar to the other joints.

There are several advantages of the reduction of maximum velocity and acceleration. From the aspect of mechanism, it decreases the stress on joints and other components of the robotic arm. While high velocity and acceleration may cause wear on motor, gear and linkages, RRT smooth algorithm is beneficial for extending the lifespan of the robotic arm through reducing these values. Moreover, a more stable motion and lower vibration is crucial for high accuracy applications such as micro-assembly tasks in the semiconductor industry. Vibration may cause errors in positioning, which results in defective products. The RRT smooth algorithm is helpful for improving the quality of products through eliminating vibration.

Meanwhile, Table IV shows the average velocity and acceleration of six joints during the motion by RRT and RRT smooth of the cube obstacle. Compared with RRT algorithm, the data generated by RRT smooth algorithm demonstrates that the average velocity of joint L4 increases from 0.017 to 0.02, while the average acceleration decreases from 0.006 to 0.002. These performances demonstrate that RRT smooth algorithm could reduce the average acceleration to decrease the vibration and wear during the motion of the robotic arm while ensuring a certain motion efficiency of joints.

Combining the data from Table III and Table IV, it is evident that the RRT smooth algorithm is more effective than RRT algorithm in improving the motion performance of the robotic arm. The RRT smooth algorithm reduces the maximum velocity, acceleration and average acceleration while increasing the motion efficiency and decrease the vibration and wear through optimizing path planning and joint motion control, which provide a powerful support for the stable operation in the complex environment of the robotic arm.

Table V and Table VI display the same parameter of the sphere obstacle as Table III and Table IV. It can be seen that the maximum velocity and acceleration generated by RRT smooth are generally smaller than RRT, which demonstrates it is more stable in the RRT algorithm. From the aspect of average velocity and acceleration RRT smooth algorithm improves the average velocity of some joints such as L1, L2 and L5 and decreases the average acceleration such as L1, L2, L3 and L5.

Table VII and Table VIII show the results based on a complex environment (the sphere and cube obstacle). Considering both spherical and cubic obstacles, the RRT smooth algorithm still demonstrates its advantages. The maximum velocity and acceleration data for each joint are lower than those of the RRT algorithm, reflecting an improvement in the stability of the robotic arm's movement. For example, at the L3 joint, the maximum speed under the RRT algorithm is 0.167 and the maximum acceleration is 0.125, while under the RRT smooth algorithm, they are 0.036 and 0.015, respectively. In terms of average speed and acceleration, the RRT smooth algorithm increases the average speed at some joints (such as L1 and L2) while reducing the average acceleration at others (such as L1, L2, L3, and L5), further proving its effectiveness in optimizing the performance of robotic arm movements.

TABLE III The maximum velocity and acceleration by RRT and RRT smooth of the cube obstacle

Joint	Algorithm			
	RRT		***RRT smooth***	
	Maximum velocity	***Maximum acceleration***	***Maximum velocity***	***Maximum acceleration***
L1	0.129	0.048	0.11	0.024
L2	0.048	0.017	0.045	0.002
L3	0.036	0.013	0.018	0.004
L4	0	0	0	0
L5	0.049	0.018	0.035	0.004
L6	0.129	0.048	0.11	0.024

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TABLE IV THE AVERAGE VELOCITY AND ACCELERATION BY RRT AND RRT SMOOTH OF THE CUBE OBSTACLE

Joint	Algorithm			
	RRT		*RRT smooth*	
	Average velocity	*Average acceleration*	*Average velocity*	*Average acceleration*
L1	0.015	0.006	0.013	0.002
L2	0.017	0.005	0.02	0.001
L3	0.008	0.003	0.005	0.001
L4	0	0	0	0
L5	0.014	0.005	0.017	0.001
L6	0.015	0.006	0.013	0.002

TABLE V THE MAXIMUM VELOCITY AND ACCELERATION BY RRT AND RRT SMOOTH OF THE SPHERE OBSTACLE

Joint	Algorithm			
	RRT		*RRT smooth*	
	Maximum velocity	*Maximum acceleration*	*Maximum velocity*	*Maximum acceleration*
L1	0.102	0.076	0.066	0.007
L2	0.099	0.074	0.062	0.008
L3	0.084	0.063	0.011	0.001
L4	0	0	0	0
L5	0.078	0.058	0.052	0
L6	0.102	0.076	0.066	0.007

TABLE VI THE AVERAGE VELOCITY AND ACCELERATION BY RRT AND RRT SMOOTH OF THE SPHERE OBSTACLE

Joint	Algorithm			
	RRT		*RRT smooth*	
	Average velocity	*Average acceleration*	*Average velocity*	*Average acceleration*
L1	0.027	0.020	0.030	0.004
L2	0.026	0.016	0.031	0.004
L3	0.019	0.015	0.005	0.001
L4	0	0	0	0
L5	0.023	0.015	0.025	0.004
L6	0.027	0.020	0.030	0.004

TABLE VII THE MAXIMUM VELOCITY AND ACCELERATION BY RRT AND RRT SMOOTH OF THE SPHERE AND CUBE OBSTACLE

Joint	Algorithm			
	RRT		*RRT smooth*	
	Maximum velocity	*Maximum acceleration*	*Maximum velocity*	*Maximum acceleration*
L1	0.079	0.060	0.070	0.028
L2	0.097	0.073	0.053	0.013
L3	0.167	0.125	0.036	0.015
L4	0	0	0	0
L5	0.078	0.059	0.019	0.002
L6	0.079	0.060	0.070	0.028

TABLE VIII THE AVERAGE VELOCITY AND ACCELERATION BY RRT AND RRT SMOOTH OF THE SPHERE AND CUBE OBSTACLE

Joint	Algorithm			
	RRT		*RRT smooth*	
	Average velocity	*Average acceleration*	*Average velocity*	*Average acceleration*
L1	0.021	0.019	0.028	0.005
L2	0.020	0.018	0.025	0.004
L3	0.026	0.023	0.017	0.003
L4	0	0	0	0
L5	0.022	0.020	0.009	0.001
L6	0.021	0.019	0.028	0.005

Fig. 2 shows the trajectory of the motion of the robotic arm as the obstacle is a cube and the yellow and green line respectively represents the trajectory generated by RRT algorithm and RRT smooth algorithm.

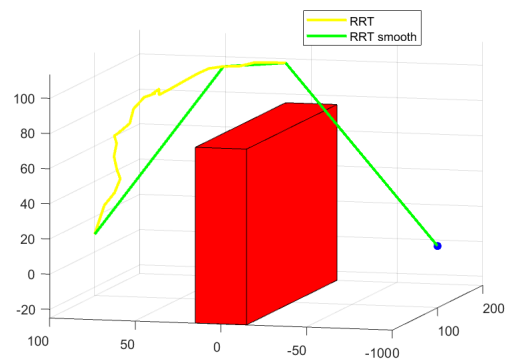

Figure 2. The trajectory of the cube obstacle

Fig. 3 and Fig. 4 show the changes of each joint angle during the motion of the robotic arm with respectively RRT and RRT smooth algorithms. Each subplot in Fig. 3 represents the angle change of a specific joint (Joint 1 to Joint 6) over time for the RRT algorithm. The x axis represents the time step, and the y axis represents the joint angle. These plots show how the angles of different joints vary during the movement of the robotic arm. For example, the angle of Joint 1 may change in a certain pattern over time, which reflects the movement trajectory of the robotic arm at that joint. However, due to the nature of the RRT algorithm, these angle changes may contain sudden fluctuations, which can be seen from the relatively sharp turns in some of the plots.

Fig. 4 shows the same information but for the RRT smooth algorithm. Comparing the two figures, it is obvious that the joint angle changes in Fig. 4 are much smoother, which owes to the use of polynomial trajectory planning in the RRT smooth algorithm as described in the methodology section. For instance, the angle change of Joint 3 in Fig. 4 is more continuous compared to that in Fig. 3. This smoothness in joint angle changes not only improves the motion stability of the robotic arm but also reduces the impact and vibration during movement, which is significant for the long-term operation and accuracy of the robotic arm. Moreover, as shown in Fig. 5, it can be seen that the obstacle avoidance is successful, while the yellow line is the trajectory generated by RRT and the green line is generated by RRT smooth when the static obstacle is changed from a cube to a sphere. The similar curves appear in Figure 6, from which obstacles are more complex combining the sphere and the cube.

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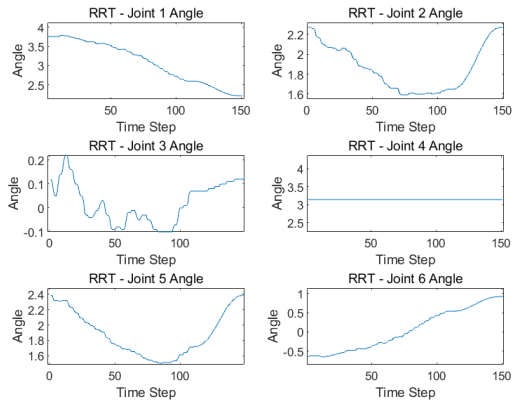

Figure 3. The angle of joints by RRT

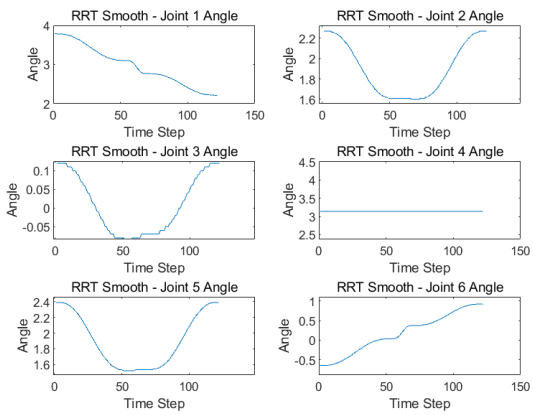

Figure 4. The angle of joints by RRT smooth

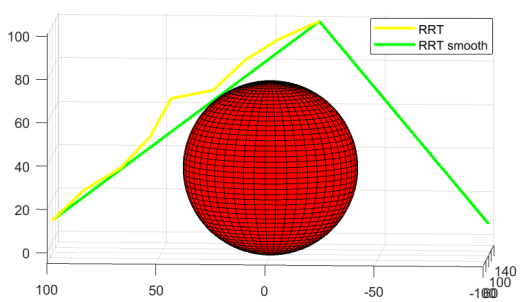

Figure 5. The trajectory of the sphere obstacle

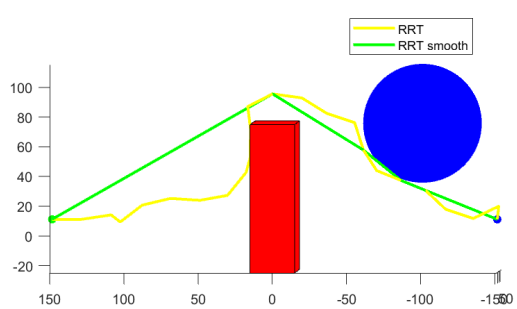

Figure 6. The trajectory of the joint of sphere and cube obstacle

To sum up, through the analysis of tables and figures, it is obvious that the RRT smooth algorithm precedes RRT algorithm from the aspect of path length, processing time and the stability of motion, which could better meet the obstacle avoidance requirement of the robotic arm. However, it is worth noting that the paths generated by both algorithms are not the optimal due to the blindness and randomness in the sampling process of the algorithm.

IV. Conclusion

The paper is dedicated to developing and exploring the obstacle avoidance algorithms for robotic arms in dynamic environment. The simulation is implemented on MATLAB and serves PUMA560 robotic arm as the research object. The simulation results demonstrate that the performance of RRT smooth algorithm is better than RRT algorithm to some extent such as the smoothness, the processing time and the velocity and acceleration of joints. Additionally, the RRT smooth algorithm optimizes the changes in joint angles during the motion of robotic arm, which makes it smoother. It could not only decrease mechanical stress and extend the lifespan of the robotic arm but also improve the overall performance of the robotic arm. However, there still exists some limitations in both algorithms due to the blindness and randomness occurring in the sampling process, which may cause that the generated path is not the shortest and optimal. Meanwhile, the efficiency of the two algorithms is limited in such a sophisticated environment, which may influence the actual time obstacle avoidance of the robotic arm.

References

[1] G. Sharma, A. S. Ganesh, S. Kumar, M. Kharub, A. Anitha and V. Kanwar, "Substantial Capabilities of 3-Axis Robotic Arm in Industry Automation Applications," *2023 3rd International Conference on Innovative Sustainable Computational Technologies (CISCT)*, Dehradun, India, 2023, pp. 1-6.

[2] P. E. Hart, N. J. Nilsson and B. Raphael, "A Formal Basis for the Heuristic Determination of Minimum Cost Paths," in *IEEE Transactions on Systems Science and Cybernetics*, vol. 4, no. 2, pp. 100-107, July 1968.

[3] O. Khatib, "Real-time obstacle avoidance for manipulators and mobile robots," *Proceedings. 1985 IEEE International Conference on Robotics and Automation*, St. Louis, MO, USA, 1985, pp. 500-505.

[4] LaValle, S. M. 1998. Rapidly - exploring random trees: A new tool for path planning. Iowa State University, Department of Computer Science, Tech. Rep. 98 – 11.

[5] Shi, W., Wang, K., Zhao, C., & Tian, M. 2022. Obstacle Avoidance Path Planning for the Dual - Arm Robot Based on an Improved RRT Algorithm. Applied Sciences, 12(8), 4087.

[6] S. Li, D. Zhao, Y. Sun, J. Yang and S. Wang, "Path Planning Algorithm Based on the Improved RRT-Connect for Home Service Robot Arms," *2021 IEEE International Conference on Intelligence and Safety for Robotics (ISR)*, Tokoname, Japan, 2021, pp. 403-407.

[7] Liu, W., Huang, Z., Qu, Y., & Chen, L. 2024. Path Planning for Robotic Arms Based on an Improved RRT Algorithm. Open Journal of Applied Sciences, 14, 1214 - 1236.

[8] Y. Zhu, Y. Tang, Y. Zhang and Y. Huang, "Path Planning of Manipulator Based on Improved RRT-Connect Algorithm," *2021 2nd International Conference on Big Data & Artificial Intelligence & Software Engineering (ICBASE)*, Zhuhai, China, 2021, pp. 44-47.

[9] Z. -J. Ding, X. -J. Meng, L. -X. Zhang, Y. -B. Guo, M. -Q. Chen and S. -H. Zhen, "Variable Sampling Area RRT Algorithm Based on Limiting Joint Angle for Robotic Arm Path Planning," in *IEEE Access*, vol. 13, pp. 30665-30677, 2025.

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